

Curriculum Vitae: Paul C. Whitford

Rice University
Center for Theoretical Biological Physics
Brockman Hall, Room 232
Houston, TX, 77005

Phone: (713) 348-4938
pwhitfor@rice.edu
<http://smog.ucsd.edu>

Education

- 2003-2009 **University of California at San Diego, San Diego, CA**
Advisor: *José N. Onuchic*
Ph.D. Physics (Biophysics): Conferred 6/13/2009
Master of Science in Physics: Conferred 3/19/2005
- 2000-2003 **Worcester Polytechnic Institute**
Bachelor of Science in Physics with *High Distinction*
Minor in Mathematics

Research Experience

- 2012-current **Research Coordinator for High Performance Computing**
Rice University
Center for Theoretical Biological Physics
- 2009-2012 **Postdoctoral Fellow**
Karissa Y. Sanbonmatsu, Los Alamos National Laboratory
Modeling ribosome structure and dynamics
- 2003-2009 **Graduate Researcher**
José N. Onuchic, University of California, San Diego, CA
Conformational transitions and folding of proteins and RNA
- 2002-2003 **Research Assistant and Undergraduate Thesis Research**
George D. J. Phillies, Worcester Polytechnic Institute, Worcester, MA
Computational investigation of glass-forming liquids
- 2001 **Research Assistant**
George D. J. Phillies, Worcester Polytechnic Institute, Worcester, MA
Light scattering spectroscopy of complex fluids

Honors and Awards

- 2009 *Director's Postdoctoral Fellowship*
Los Alamos National Laboratory
- 2009 *Funded Visits for Outstanding Students Award*
Weizmann Institute of Science, Israel
- 2004-2009 *Center for Theoretical Biological Physics Fellow*
University of California at San Diego
- 2007 *International ICAM Junior Exchange Award*
Funded collaboration with scientists in São Paulo, Brazil

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- 2005-2007 *Molecular Biophysics Training Grant Fellow*
University of California at San Diego
- 2003-2005 *San Diego Fellowship*
Office of Graduate Studies, University of California at San Diego
- 2003 *SPS Leadership Scholarship*
Awarded by the Society of Physics Students national organization
- 2003 *Dr. Robert H. Goddard Award* for outstanding performance in research
Faculty of Physics, Worcester Polytechnic Institute, Worcester, MA
- 2002 Inducted into *Tau Beta Pi*: National Engineering Honor Society
- 2002 Inducted into *Pi Mu Epsilon*: National Mathematics Honor Society
- 2002 Inducted into *Sigma Pi Sigma*: National Physics Honor Society

Invited Talks

- [24] State University of New York at Albany. Department of Chemistry. Albany, NY. *March 7th, 2012.*
- [23] Biophysical Society 56th Annual Meeting. San Diego, CA. *February 26th, 2012.*
- [22] Hospital for Sick Children. Molecular Structure and Function Program seminar series. Toronto, Canada. *October 24th, 2011.*
- [21] 1st KIAS Conference on Subcellular Dynamics. Seoul, South Korea. *July 26th, 2011.*
- [20] International Conference on Biological Physics Workshop. San Diego, CA. *June 21st, 2011.*
- [19] University of Rochester Medical Center. Department of Biochemistry and Biophysics. Rochester, NY. *March 10th, 2011.*
- [18] College of Staten Island. Department of Chemistry. New York, NY. *November 18th, 2010.*
- [17] State University of New York at Stony Brook. Department of Chemistry. Stony Brook, NY. *November 15th, 2010.*
- [16] City College of New York. Department of Physics. New York, NY. *November 10th, 2010.*
- [15] Hebrew University in Jerusalem. The Alexander Silberman Institute of Life Sciences. Jerusalem, Israel. *August 16th, 2010.*
- [14] Weizmann Institute of Science. Department of Structural Biology. Rehovot, Israel. *July 21st, 2010.*
- [13] TSRI Characterizing Energy Landscapes Workshop. Telluride, CO. *June 15th,*

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2010.

[12] University of Potsdam. Institute of Biochemistry and Biology. Potsdam, Germany. *April 29th, 2010.*

[11] Charité Universitätsmedizin Berlin. Institut Für Medizinische Physik und Biophysik. Berlin, Germany. *February 15th, 2010.*

[10] Max Planck Institute for Biophysical Chemistry. Department of Theoretical and Computational Chemistry. Göttingen, Germany. *February 10th, 2010.*

[9] 23rd tRNA Workshop. Aveiro, Portugal. *February 1st, 2010.*

[8] University of California at Santa Cruz. Department of Molecular, Cellular and Developmental Biology. Santa Cruz, CA. *September 21st, 2009.*

[7] RNA Society 14th Annual Meeting. Madison, Wisconsin, *May 28th, 2009.*

[6] Guest Lecture. CSU San Marcos. Department of Physics. San Marcos, CA. *March 16th, 2009.*

[5] Weill Medical College of Cornell University. Department of Physiology and Biophysics. New York, NY. *March 9th, 2009.*

[4] Rensselaer Polytechnic Institute. Department of Physics. Troy, NY. *March 5th, 2009.*

[3] Biophysical Society 53rd Annual Meeting. Boston, MA. *March 4th, 2009.*

[2] Universidade Estadual Paulista. Department of Physics. São José do Rio Preto, São Paulo, Brazil. *September 27th, 2007.*

[1] TSRI Energy Landscapes Workshop. Telluride, CO. *April 4th, 2007.*

Reviewer Activity

Proceedings of the National Academy of Sciences
PNAS Plus

Journal of the American Chemical Society

Biophysical Journal

Proteins: Structure, Function and Bioinformatics

Journal of Physical Chemistry

PLoS Computational Biology

PLoS ONE

Molecular BioSystems

Biopolymers

BMC Bioinformatics

RNA

Chemical Physics Letters

Pacific Symposium on Biocomputing proceedings

National Science Foundation

Partnership for Advanced Computing in Europe (PRACE)

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Awarded Computing Allocation

1/1/11-12/31/11: TeraGrid Computing Award for “Using High Performance Computing to Reveal the Functional Dynamics of the Ribosome.” 4.1 million Service Units on the AQS, Trestles and Lonestar clusters (1 Service Unit ~ 1 core-hour). Role: PI.

PDB Entries

2XSY, 2XTG, 2XUX, 2XUY: Atomic models of the 70S ribosome, based on cryo-EM densities. Prepared with MDfit (See Ratje et al. *Nature* 2010).

Teaching/Mentoring

- May 2007 **Principal Organizer and Presenter** CTBP-Workshop on Protein Dynamics, *Models, Simulations and Thermodynamics* workshop for graduate students
- 2006 **Mentor** of Research Experience for Undergraduate (REU) student.
- 2004 **Organizer and Presenter:** “Physics Majors Users Guide (PMUG)”
This seminar series exposed undergraduates to many areas of physics.
Department of Physics, University of California at San Diego
- 2003-2005 **Participant** Preparing Future Physics Faculty Program
Department of Physics, University of California at San Diego
- 2001-2003 **Peer Learning Assistant (Undergraduate TA)**
Department of Mathematics
Worcester Polytechnic Institute, Worcester, MA

Service

- 2003-2006 *UCSD Graduate Student Association*
Vice-president of External Affairs in 2004-2005
Graduate representative to the state and federal legislature
- 2004-2005 *University of California Student Association*
Chair of Graduate and Professional Student Committee
Graduate representative to the state and federal legislature and UCOP
- 2004-2005 *Science Policy Analysis Roundtable*
Co-organizer and participant
Activities included journal club-like seminars with graduate student, post-doctoral and faculty speakers in addition to large forums with invited speakers from across California.
- 2004-2009 *UCSD Academic Integrity Hearing Board*
Board Member
- 2000-2003 *Society of Physics Students*
President for 2002-2003 of Worcester Polytechnic Institute Chapter
Member of National Council, representing New England for 2002-03
Principal organizer of New England Zone meeting

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Research Articles (* shared authorship, ^corresponding authorship)

- [22] R. L. Hayes, J. K. Noel, U. Mohanty, P. C. Whitford, S. P. Hennelly, J. N. Onuchic, K. Y. Sanbonmatsu. Magnesium fluctuations modulate RNA dynamics in the SAM-1 riboswitch. *J. Amer. Chem. Soc.* In Press.
- [21] J. K. Noel, P. C. Whitford and J. N. Onuchic. The shadow method: A Generally-Applicable contact definition for capturing the dynamics of biomolecular folding and function. *J. Phys. Chem.* 2012. In Press.
- [20] J. Singh, P. C. Whitford, N. Hayre, J. N. Onuchic and D. Cox. Massive conformation change in the prion protein: Using dual-basin structure-based models to find misfolding pathways. *Proteins: Struct. Func. Bioinfo.* DOI:10.1002/prot.24026
- [19] A. Ahmed, P. C. Whitford, K. Y. Sanbonmatsu and F. Tama. Consensus between flexible fitting computational methods to interpret cryo-EM data. *J. Struct. Biol.* **177**, 561-570, 2012.
- [18] P. C. Whitford, A. Ahmed, Y. Yu, S. P. Hennelly, F. Tama, C. M. T. Spahn, J. N. Onuchic and K. Y. Sanbonmatsu. Excited states of ribosome translocation revealed through integrative molecular modeling. *Proc. Nat. Acad. Sci. USA.* **108**, 18943-18948, 2011.
- [17] R. Nechushtai, H. Lammert, D. Michaeli, Y. Eizenberg-Domovish, J. A. Zuris, M. A. Luca, D. T. Capraro, A. Fish, O. Shimshon, M. Roy, A. Schug, P. C. Whitford, O. Livnah, J. N. Onuchic and P. A. Jennings. Allostery in the ferredoxin protein motif does not involve a conformational switch. *Proc. Nat. Acad. Sci. USA.* **108**, 2240-2245, 2011.
- [16] P. C. Whitford^, J. N. Onuchic and K. Y. Sanbonmatsu. Connecting energy landscapes and experimental rates for aminoacyl-tRNA accommodation in the ribosome. *J. Amer. Chem. Soc.* **132**, 13170-13171, 2010. (communication)
- [15] A. H. Ratje, J. Loerke, A. Mikolajka, M. Br  nner, P. W. Hildebrand, A. Starosta, A. Doenhoefer, S. R. Connell, P. Fucini, T. Mielke, P. C. Whitford, J. N. Onuchic, Y. Yanan, K. Y. Sanbonmatsu, R. K. Hartmann, P. A. Penczek, D. N. Wilson and C. M. T. Spahn. Head swivel on the ribosome facilitates translocation via intra-subunit tRNA hybrid sites. *Nature* **468**, 713-716, 2010.
- [14] M. A. Jamros, L. C. deOliveira, P. C. Whitford, J. N. Onuchic, J. A. Adams, D. K. Blumenthal and P. A. Jennings. Proteins at work: A combined SAXS and theoretical determination of the multiple structures involved on the protein kinase functional landscape. *J. Biol. Chem.* **285**, 36121-36128, 2010.
- [13] J. K. Noel, P. C. Whitford, K. Y. Sanbonmatsu, and J. N. Onuchic. SMOG@ctbp: Simplified deployment of structure-based models in GROMACS. *Nucleic Acid Research* **38**(suppl. 2), W657-W661, 2010.
- [12] P. C. Whitford, P. Geggier, R. Altman, S. C. Blanchard, J. N. Onuchic, and K. Y. Sanbonmatsu. Accommodation of aminoacyl-tRNA into the ribosome involves reversible excursions along multiple pathways. *RNA.* **16**, 1196-1204, 2010.

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- [11] R. J. Oliveira*, P. C. Whitford*[^], J. Chahine, J. Wang, J. N. Onuchic, and V. B. P. Leite[^]. Exploring the origin of non-monotonic complex behavior and the effects of non-native interactions on the diffusive properties of protein folding. *Biophys. J.* **99**, 600-608, 2010.
- [10] P. C. Whitford, A. Schug, J. Saunders, S. P. Hennelly, J. N. Onuchic, and K. Y. Sanbonmatsu. Nonlocal helix formation is key to understanding S-adenosylmethionine riboswitch function. *Biophys. J.* **96**, L7-9, 2009. (On cover).
- [9] P. C. Whitford, J. K. Noel, S. Gosavi, A. Schug, K. Y. Sanbonmatsu, and J. N. Onuchic. An all-atom structure-based potential for proteins: Bridging minimal models with all-atom empirical forcefields. *Prot. Struct. Func. Bioinfo.* **75**, 430-441, 2009.
- [8] S. Gosavi, P. C. Whitford, P. A. Jennings, and J. N. Onuchic. Extracting function from a β -trefoil folding motif. *Proc. Nat. Acad. Sci.* **105**, 10384–10389, 2008.
- [7] P. C. Whitford, S. Gosavi, and J. N. Onuchic. Conformational transitions in adenylate kinase - Allosteric communication reduces misligation. *J. Biol. Chem.* **283**, 2042–2048, 2008.
- [6] A. Schug, P. C. Whitford, Y. Levy, and J. N. Onuchic. Mutations as trapdoors to two competing native conformations of the rop-dimer. *Proc. Nat. Acad. Sci.* **104**, 17674–17679, 2007.
- [5] P. C. Whitford, O. Miyashita, Y. Levy, and J. N. Onuchic. Conformational transitions of adenylate kinase: Switching by cracking. *J. Mol. Biol.* **366**, 1661–1671, 2007.
- [4] J. E. Mills, P. C. Whitford, J. Shaffer, J. N. Onuchic, J. A. Adams, and P. A. Jennings. A novel disulfide bond in the SH2 domain of the C-terminal Src kinase controls catalytic activity. *J. Mol. Biol.* **365**, 1460–1468, 2007.
- [3] P. C. Whitford, and G. D. J. Phillies. Enhanced septahedral ordering in cold Lennard-Jones fluids. *Phys. Rev. E* **72**, 021203, 2005.
- [2] P. C. Whitford, and G. D. J. Phillies. Extended-range order, diverging static length scales, and local structure formation in cold Lennard-Jones fluids. *J. Chem. Phys.* **122**, 044508, 2005.
- [1] G. D. J. Phillies, R. O'Connell, P. Whitford, and K. A. Streletzky. Mode structure of diffusive transport in hydroxypropylcellulose:water. *J. Chem. Phys.* **119**, 9903–9913, 2003.

Review Articles

- [2] P. C. Whitford, K. Y. Sanbonmatsu, J. N. Onuchic. Biomolecular dynamics: Order-disorder transitions and energy landscapes. *Rep. Prog. Phys.* 2012. In Press.
- [1] R. J. Oliveira, P. C. Whitford, J. Chahine, V. B. P. Leite and J. Wang. Coordinate and time dependent diffusion dynamics in protein folding. *Methods* **52**, 91-98, 2010.

Commentary

P. C. Whitford, J. N. Onuchic, and P. G. Wolynes. Energy landscape along an enzymatic

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reaction trajectory: hinges or cracks? *HFSP J.* **2**, 61–64, 2008.

Book Chapters

[4] K. Y. Sanbonmatsu, S. C. Blanchard and P. C. Whitford. Dynamics of very large systems: The Ribosome. *Comprehensive Biophysics*. 2012. In Press.

[3] K. Y. Sanbonmatsu, S. C. Blanchard and P. C. Whitford. Molecular dynamics simulations of the ribosome. *Biophysical approaches to translational control of gene expression*. 2012. ISBN: 978-1-4614-3990-5

[2] P. C. Whitford, R. B. Altman, P. Geggier, D. Terry, J. B. Munro, J. N. Onuchic, C. M. T. Spahn, K. Y. Sanbonmatsu, S. C. Blanchard. Chapter 24: Dynamic views of ribosome function: energy landscapes and ensembles. *The Ribosome: Structure, Function, and Dynamics*. 2011. ISBN: 978-3-7091-0214-5.

[1] K. Y. Sanbonmatsu, S. C. Blanchard and P. C. Whitford. Chapter 4: Information processing by nanomachines: decoding by the ribosome. *Molecular Machines*. 2011. ISBN-13: 978-98143442

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